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Metric Structures for Riemannian and Non-Riemannian Spaces

Misha Gromov

With Appendices by
M. Katz, P. Pansu, and S. Semmes

English translation
by Sean Michael Bates

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...Meme ceux qui furent favorables à ma perception des vérités que je voulais ensuite graver dans le temple, me félicitèrent de les avoir découvertes au microscope, quand je m'étais au contraire servi d'un télescope pour apercevoir des choses, très petites en effet, mais parce qu'elles étaient situées à une grande distance, et qui étaient chacune un monde.

Marcel Proust, *Le temps retrouvé*
(Pléiade, Paris, 1954, p. 1041)

Contents

Preface to the French Edition

This book arose from a course given at the University of Paris VII during the third trimester of 1979. My purpose was to describe some of the connections between the curvature of a Riemannian manifold V and some of its global properties. Here, the word *global* refers not only to the topology of V but also to a family of metric invariants, defined for Riemannian manifolds and mappings between these spaces. The simplest metric invariants of such V are, for example, its volume and diameter; an important invariant of a mapping from V_1 into V_2 is its dilatation. In fact, such invariants also appear in a purely topological context and provide an important link between the given infinitesimal information about V (usually expressed as some restriction on the curvature) and the topology of V.

For example, the classical Gauss–Bonnett theorem gives an upper bound for the diameter of a positively curved manifold V, from which one can deduce the finitude of the fundamental group of V. For a deeper topological study of Riemannian manifolds, we need more subtle invariants than diameter or volume; I have attempted to present a systematic treatment of these invariants, but this treatment is far from exhaustive.

Messrs. J. Lafontaine and P. Pansu have successfully completed the almost insurmountable task of transforming into a rigorous mathematical text the chaos of my course, which was scattered with imprecise statements and incomplete proofs. I thank them as well as M. Berger, without whose assistance and encouragement this book would never have come into being. I also thank the *Editions Cedic* for the liberty it afforded the authors at the time that the book was edited.

M. Gromov
Paris, June 1980

Preface to the English Edition

The metric theory described in this book covers a domain stretching somewhere between the fields of topology and global Riemannian geometry. The boundary of this domain has dramatically exploded since 1979 and accordingly, in the course of its translation from the 1979 French version into English, the book has approximately quadrupled in size, even though I tried not to maim the original text with unnecessary incisions, insertions, and corrections, but rather to add several new sections indicated by the + subscript. The most voluminous additions are Chapter $3\frac{1}{2}_+$, which attempts to link geometry and probability theory, and Appendix B_+, where an analyst lays down his view on metric spaces. Here, the reader can painlessly learn several key ideas of real analysis made accessible to us geometers by the masterful exposition of Stephen Semmes, who has adapted his material to our non-analytic minds.

Additionally, Appendix C_+ reproduces my 1980-rendition of Paul Levy's inequality, while Misha Katz gives an overview of systolic freedom in Appendix D_+.

Acknowledgements: The initiative to publish an English translation of the book with Birkhäuser is due to Alan Weinstein. It was my pleasure to cooperate with Sean Bates, who translated the original version of the book and helped me edit the new sections. I am also grateful to Marcel Berger, Keith Burns, and Richard Montgomery for calling my attention to errors in the first version of the book.

M. Gromov
Bures-sur-Yvette, May 1997

Introduction: Metrics Everywhere

The conception of "distance" is already present in everyday language where it refers to two physical objects or two abstract ideas being mutually close or far apart. The most common (but by no means most general) mathematical incarnation of this vague idea is the notion of *metric space*, that is, an abstract set X where the distance between its elements, called *points* $x \in X$, is measured by *positive real numbers.* Thus a metric space is a set X with a given function in two variables, say $d : X \times X \rightarrow \mathbb{R}_+$ satisfying the famous *triangle inequality*

$$d(x, x'') \leq d(x, x') + d(x', x'')$$

for all triples of points x, x' and x'' in X.

Besides, one insists that the distance function be *symmetric*, that is, $d(x, x') = d(x', x)$. (This unpleasantly limits many applications: the effort of climbing up to the top of a mountain, in real life as well as in mathematics, is not at all the same as descending back to the starting point).

Finally, one assumes $d(x, x) = 0$ for all $x \in X$ and add the following *separation axiom.* If $x \neq x'$, then $d(x, x') = 0$. This seems to be an innocuous restriction, as one can always pass to the quotient space by identifying x and x' whenever $d(x, x') = 0$. But sometimes the separation becomes a central issue, e.g., for *Kobayashi* and *Hofer metrics*, where such identification may reduce X to a single point, for instance).

The archtypical example of a metric space is the ordinary Euclidean space $\mathbb{R}^n$ with the pythagorean distance between the points $x = (x_1, \ldots, x_n)$ and $x' = (x'_1, \ldots, x'_n)$ defined by

$$d(x, x') = \sqrt{(x_1 - x'_1)^2 + \cdots + (x_n - x'_n)^2}.$$

Next come subsets in $\mathbb{R}^n$ with this metric providing many appealing examples, such as *the sphere* $S^{n-1} = \{x \in \mathbb{R}^n \,|\, \sum_{i=1}^n x_i^2 = 1\}$ and the set of vertices of the unit cube, i.e., $\{0,1\}^n \subset \mathbb{R}^n$ with the *induced Euclidean* (pythagorean) distance. If X is a smooth connected submanifold in $\mathbb{R}^n$ (as the above sphere) then, besides the induced Euclidean distance $\mathrm{dist}_{\mathbb{R}^n}$ on X, one has the *induced Riemannian distance*, $\mathrm{dist}_X(x, x')$ defined as the infimum of lengths of curves *contained in* X and joining x and x'. (One may be tempted to use the above as a quick definition of a Riemannian metric. Indeed, every Riemannian manifold admits a smooth embedding into some $\mathbb{R}^n$ preserving the length of the curves according to the Nash theorem. But Euclidean embeddings hide rather than reveal the true metric structure of Riemannian manifolds due to uncontrolled distortion $\mathrm{dist}_X \mid \mathrm{dist}_{\mathbb{R}^n}$. Besides the full beauty and power of Riemannian geometry depend not only on the metric but also on the associated elliptic Riemannian equations, such as Laplace–Hodge, Dirac, Yang–Mills, and so on. These naturally come along with the Riemannian tensor but are nearly invisible on X embedded to $\mathbb{R}^n$.

Our approach to general metric spaces bears the undeniable imprint of early exposure to Euclidean geometry. We just love spaces sharing a common feature with $\mathbb{R}^n$. Thus there is a long tradition of the study of *homogeneous* spaces X where the isometry group acts *transitively* on X. (In the Riemannian case the metric on such X is fully determined by prescribing a positive definite quadratic form g_o on a single tangent space $T_{x_o}(X)$. But the simplicity of this description is illusory; it is quite hard to evaluate metric invariants of X in terms of g_o. For example, one has a very limited idea of how *systoles* (see below) behave as one varies a left invariant metric on a Lie group $SO(n)$ or $U(n)$ for instance.) Besides isometries, $\mathbb{R}^n$ possesses many nontrivial *self-similarities*, i.e., transformations f, such that $f^*(\mathrm{dist}) = \lambda$ dist for some constant $\lambda \neq 0, 1$. There are no self-similar spaces besides $\mathbb{R}^n$ in the Riemannian category — this is obvious — but there are many such non-Riemannian examples such as p-adic vector spaces (these are totally disconnected) and some connected nilpotent Lie groups (e.g., the *Heisenberg group*) with Carnot–Carathéodory metrics (see 1.4, 1.18 and 2.6 in Appendix B).

Switching the mental wavelength, one introduces spaces with curvature $K \leq 0$ and $K \geq 0$ by requiring their small geodesic triangles to be "thinner" (correspondingly, "thicker") than the Euclidean ones (see 1.19_+). Here one is guided by the geometry of *symmetric spaces* that are distinguished homogeneous spaces, such as S^n and $\mathbb{C}P^n$ where $K \geq 0$ and $SL_n\mathbb{R}/SO(n)$ with $K \leq 0$.

Apart from direct Euclidean descendants there are many instances of metrics associated to various structures, sometimes in a rather unexpected and subtle way. Here are a few examples.

Complex manifolds. The complex space $\mathbb{C}^n, n \geq 1$, carries no metric invariant under holomorphic automorphisms. There are just too many of them! Yet, many complex (and almost complex) manifolds, e.g., *bounded* domains in $\mathbb{C}^n$, do possess such natural metrics, for example, the *Kobayashi metric* (see 1.8 bis_+).

Symplectic manifolds. No such manifold X of positive dimension carries an invariant metric, again because the group of symplectomorphisms is too large. Yet, the infinite dimensional space of closed *lagrangian* submanifolds in X (or rather each "hamiltonian component" of this space) does admit such a metric. (The construction of the metric is easy but the proof of the separation property, due to Hofer, is quite profound. Alas, we have no room for Hofer's metric in our book).

Homotopy category. Once can functorially associate an infinite dimensional metric polyhedron to the homotopy class of each topological space X, such that continuous maps between spaces transform to distance decreasing maps between these polyhedra. Amazingly, the metric invariants of such polyhedra (e.g., its systoles, the volumes of minimal subvarieties realizing prescribed homology classes) lead to new homotopy invariants which are most useful for (aspherical) spaces X with large fundamental groups (see Ch. 5H_+).

Discrete groups. A group with a distinguished finite set of generators carries a natural discrete metric which only moderately, i.e., bi-Lipschitzly, changes with a change of generators. Then, by adopting ideas from the geometry of noncompact Riemannian manifolds, one defines a variety of asymptotic invariants of infinite groups that shed new light on the whole body of group theory (see 3C, 5B, and 6B, C).

Lipschitz and bi-Lipschitz. What are essential maps between metric spaces? The answer "isometric" leads to a poor and rather boring category. The most generous response "continuous" takes us out of geometry to the realm of pure topology. We mediate between the two extremes by emphasizing *distance decreasing maps* $f : X \to Y$ as well as general *λ-Lipschitz* maps f which enlarge distances at most by a factor λ for some $\lambda \geq 0$.

Isomorphisms in this categeory are *λ-bi-Lipschitz homeomorphisms* and most metric invariants defined in our book transform in a λ-controlled way under λ-Lipschitz maps, as does for example, the *diameter* of a space, $\text{Diam}X = \sup_{x,x'} \text{dist}(x, x')$. We study several classes of such invariants with a special treatment of *systoles* measuring the minimal volumes of homology classes in X (see Ch. 4 and App. D) and of *isoperimetric profiles* of complete Riemannian manifolds and infinite groups which are linked in

Ch. 6 to quasiconformal and quasiregular mappings.

Asymptotic viewpoint. Since every diffeomorphism between *compact* Riemannian manifolds is λ-bi-Lipschitz for some $\lambda < \infty$, our invariants tell us preciously little if we look at a *fixed compact* manifold X. What is truly interesting in the λ-Lipschitz environment is the metric behavior of *sequences* of compact spaces. This ideology applies, for example, to an individual *noncompact* space X, where the asymptotic geometry may be seen as X is exhausted by a growing sequence of compact subspaces. We also study sequences of maps and homotopy classes of maps between fixed compact spaces, say $f_i : X \to Y$ (see Ch. 2, 7) and relate this asymptotic metric behavior to the homotopy structure of X and Y (with many fundamental questions remaining open).

Metric sociology. As our perspective shifts from individual spaces X to families (e.g., sequences) of these, we start looking at all metric spaces simultaneously and observe that there are several satisfactory notions of distance *between* metric spaces (see Ch. 3). Thus we may speak of various kinds of metric convergence of a sequence X_i to a space X and study the asymptotics of particular sequences. For example, if X_i, $i = 1, 2, \dots$, are Riemannian manifolds of dimension n with a fixed bound on their sectional curvatures, then there is a subsequence that converges (or collapses) to a mildly singular space of dimension $m \leq n$ (see Ch. 8).

Metric, Measure and Probability. Suppose our metric spaces are additionally given some measures, e.g., the standard Riemannian measures if we deal with compact Riemannian manifolds. Here one has several notions of metric convergence of spaces modulo subsets of measure $\to 0$ (see Ch. $3\frac{1}{2}{}_+$). Then there is a weaker convergence most suitable for sequences X_i with $\dim X_i \to \infty$. According to this, unit spheres $S^i \subset \mathbb{R}^{i+1}$ with normalized Riemannian measures converge (or concentrate, see $3\frac{1}{2}{}_+$) to a single atom of unit mass! This can be regarded as a geometric version of the *law of large numbers* that is derived in the present case from the *spherical isoperimetric inequality* (see $3\frac{1}{2}{}_+$E and Appendix C).

From local to global. This is a guiding principle in geometry as well as in much of analysis, and the reader will find it in all corners of our book. It appears most clearly in Ch. 5 where we explain how the lower bound on the *Ricci curvature of a manifold* X implies the *measure doubling property*, saying that the volume of each not very large $2r$-ball $B(2r) \subset X$ does not exceed const Vol $B(r)$ for the concentric ball $B(r) \subset B(2r)$. This leads to several topological consequences concerning the fundamental group of X (see Ch. 5).

Besides the volumes of balls, the Ricci curvature controls the isoperimetric profile of X. For example, the spherical isoperimetric inequality generalizes to the manifolds with Ricci $\geq -$ const (see Appendix C).

Analysis on metric spaces. Smooth manifolds and maps, being infinitesimally linear, appear plain and uneventful when looked upon through a microscope. But singular fractal spaces and maps display a kaleidoscopic variety of patterns on all local scales. Some of these spaces and maps are sufficiently regular, e.g., they may possess the doubling property, and provide a fertile soil for developing rich geometric analysis. This is exposed by Stephen Semmes in Appendix B.

I have completed describing what is in the book. It would take another volume just to indicate the full range of applications of the metric idea in various domains of mathematics.

Misha Gromov

April 1999

Chapter 1

Length Structures: Path Metric Spaces

Introduction

In classical Riemannian geometry, one begins with a C^∞ manifold X and then studies smooth, positive-definite sections g of the bundle S^2T^*X. In order to introduce the fundamental notions of covariant derivative and curvature (cf. [Grl–Kl–Mey] or [Milnor]$_{\mathrm{MT}}$, Ch. 2), use is made only of the differentiability of g and not of its positivity, as illustrated by Lorentzian geometry in general relativity. By contrast, the concepts of the length of curves in X and of the geodesic distance associated with the metric g rely only on the fact that g gives rise to a family of continuous norms on the tangent spaces T_xX of X. We will study the associated notions of length and distance for their own sake.

A. Length structures

1.1. Definition: The *dilatation* of a mapping f between metric spaces X, Y is the (possibly infinite) number

$$\mathrm{dil}(f) = \sup_{\substack{x,x' \in X \\ x \neq x'}} \frac{d(f(x), f(x'))}{d(x,x')},$$

where "d" stands for the metrics (distances) dist_X in X and dist_Y in Y. The *local dilatation* of f at x is the number

$$\mathrm{dil}_x(f) = \lim_{\varepsilon \to 0} \mathrm{dil}\left(f|_{B(x,\varepsilon)}\right).$$

A map f is called *Lipschitz* if $\mathrm{dil}(f) < \infty$; it is called λ-*Lipschitz* if $\mathrm{dil}(f) \leq \lambda$, in which case, the infimal such λ is called *the Lipschitz constant* of f.

If f is a Lipschitz mapping of an interval $[a, b]$ into X, then the function $t \mapsto \mathrm{dil}_t(f)$ is measurable.

1.2. Definition: The *length* of a Lipschitz map $f : [a, b] \to X$ is the number

$$\ell(f) = \int_a^b \mathrm{dil}_t(f)\, dt.$$

If f is merely continuous, we can define $\ell(f)$ as the supremum of all sums of the form $\sum_{i=0}^n d(f(t_i), f(t_{i+1}))$ where $a = t_0 \leq t_1 \leq \cdots \leq t_{n+1} = b$ is a finite partition of $[a, b]$.

If φ is a homeomorphism of a closed interval I' onto $I = [a, b]$, then ℓ satisfies $\ell(f \circ \varphi) = \ell(f)$, as follows from the fact that φ is strictly monotone (invariance under change of parameter).

The two definitions of $\ell(f)$ stated above are equivalent when f is absolutely continuous (cf. [Rinow], p. 106). This fact permits us to define $\ell(f)$ as the integral of the local dilatation when f is Lipschitz and to set $\ell(f \circ \varphi) = \ell(f)$ for each homeomorphism φ of I onto I'. More generally,

1.3. Definition: A *length structure* on a set X consists of a family $\mathcal{C}(I)$ of mappings $f : I \to X$ for each interval I and a map ℓ of $\mathcal{C} = \bigcup \mathcal{C}(I)$ into $\mathbb{R}$ having the following properties:

(*a*) **Positivity:** We have $\ell(f) \geq 0$ for each $f \in \mathcal{C}$, and $\ell(f) = 0$ if and only if f is constant (we assume of course that the constant functions belong to $\mathcal{C}$).

(*b*) **Restriction, juxtaposition:** If $I \subset J$, then the restriction to I of any member of $\mathcal{C}(J)$ is contained in $\mathcal{C}(I)$. If $f \in \mathcal{C}([a, b])$ and $g \in \mathcal{C}([b, c])$, then the function h obtained by juxtaposition of f and g lies in $\mathcal{C}([a, c])$ and $\ell(h) = \ell(f) + \ell(g)$.

(*c*) **Invariance under change of parameter:** If φ is a homeomorphism from I onto J and if $f \in \mathcal{C}(J)$, then $f \circ \varphi \in \mathcal{C}(I)$ and $\ell(f \circ \varphi) = \ell(f)$.

(*d*) **Continuity:** For each $I = [a, b]$, the map $t \mapsto \ell(f|_{[a,t]})$ is continuous.

Using conditions $(a), (b)$, and (c), we can define a pseudo-metric d_ℓ on X called the *length metric* by setting

$$d_\ell(x, y) = \inf\{\ell(f) : f \in \mathcal{C}, x, y \in \mathrm{im}(f)\}.$$

As usual, this pseudo-metric induces a topology on X.

It is common to define $\ell(f) = \infty$ when the map $f : I \to X$ is not contained in $C(I)$.

1.4. Examples:

(a) A metric space (X, d) is equipped with a canonical length structure: The set C consists of all continuous mappings from intervals into X, and the function ℓ is defined as in 1.2 above. The resulting structure is called the *metric length structure* of (X, d); in general, however, the length metric d_ℓ differs from d, and their corresponding topologies may also be distinct.

(b_+) *Tits-like metrics and snowflakes*: Consider $\mathbb{R}^n$ equipped with polar coordinates (r, s), where $r \in [0, \infty)$ and $s \in S^{n-1}$, the unit sphere in $\mathbb{R}^n$. Define

$$d(x_1, x_2) = |r_1 - r_2| + r\|s_1 - s_2\|^{1/2},$$

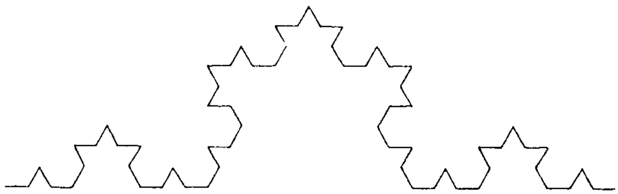
where $x_i = (r_i, s_i) \in \mathbb{R}^n$, $i = 1, 2$, $\|s_1 - s_2\|$ denotes the Euclidean distance on $S^{n-1} \subset \mathbb{R}^n$, and $r = \min\{r_1, r_2\}$. This d gives rise to the usual topology on $\mathbb{R}^n$, but

$$d_\ell(x_1, x_2) = \begin{cases} |r_1 - r_2| & \text{for } s_1 = s_2 \\ r_1 + r_2 & \text{for } s_1 \neq s_2, \end{cases}$$

and so $(\mathbb{R}^n, d_\ell)$ becomes the *disjoint* union of the Euclidean rays $[0, \infty) \times s$ for all $s \in S^{n-1}$, all glued together at the origin only. In particular, the unit sphere $S^{n-1} \subset \mathbb{R}^n$ is discrete with respect to d_ℓ. Metrics of this type naturally appear on (the ideal boundaries of) manifolds with nonpositive sectional curvatures and are collectively referred to as *Tits metrics* (cf. [Ba–Gr–Sch]).

An analog of the metric d_ℓ can be constructed on the subset of Euclidean 3-space consisting of the straight cone $X \subset \mathbb{R}^3$ over the Koch snowflake $S \subset \mathbb{R}^2$. (Here, the snowflake is the base of the cone and plays the role of the sphere S^{n-1} in the Tits-like example above). The only curves in X having finite Euclidean length are those contained in the (straight) generating lines of the cone, and so these lines are disjoint with respect to d_ℓ away from the vertex (compare [Rinow], p. 117, and Appendix B_+ of this book).

In general, the metrics d, d_ℓ always satisfy the inequality $d \leq d_\ell$, so that their corresponding topologies coincide if and only if for each $x \in X$ and $\varepsilon > 0$, there exists a d-neighborhood of x in which each point is connected to x by a curve of length at most ε.

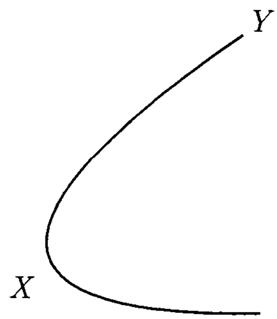


(*c*) If X is a manifold, then any Riemannian or Finslerian structure on X naturally gives rise to a length structure: One proceeds as in 1.2, noting that when f is differentiable, its local dilatation at a point x equals the norm of its derivative at x.

(*d*) *Induced length structures*: If X is equipped with a length structure and φ is a map from a set Y into X, then we obtain a length structure on Y by setting

$$\ell_Y(f) = \ell_X(\varphi \circ f)$$

for each $f\colon I \to Y$.



(*e*) *First exposure to Carnot–Caratheodory spaces.* We can associate a length structure on a Riemannian manifold (V, g) with any tangent subbundle $E \subset TV$ by defining the length of a curve c to be its usual Riemannian length if c is absolutely continuous and its tangent vector lies within E at a.e. point, and by setting $\ell(c) = \infty$ otherwise. If E is integrable, then the topology defined by d_ℓ is none other than the leaf topology. The case of nonintegrable E is of great interest.

A basic example of the latter structure is provided by the 3-dimensional Heisenberg group $\mathbb{H}^3$ of matrices of the form

$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix},$$

equipped with a left-invariant metric. The quotient of $\mathbb{H}^3$ by its center C (isomorphic to $\mathbb{R}$) defines a Riemannian fibration (see [Ber–Gau–Maz], Ch. 1) of $\mathbb{H}^3$ over the Euclidean plane $\mathbb{H}^3/C \simeq \mathbb{R}^2$. The subbundle E then consists of the horizontal subbundle of this fibration, which coincides with the kernel of the 1-form $dz - x\,dy$.

1.5. Suppose X is equipped with a length structure ℓ, and let $\tilde{\ell}$ be the length structure defined by the metric d_ℓ. The following criterion, which is nothing more than an axiomatic version of the classical properties of the lengths of curves in metric spaces, describes when these two structures are identical.

1.6. Proposition: *If, for each interval I, the function ℓ is lower semicontinuous on $C(I)$ with respect to the compact-open topology, then $\ell = \tilde{\ell}$.*

Proof. By 1.3(d), the function $t \mapsto \ell(f|_{[a,t]})$ is uniformly continuous on $I = [a, b]$. For each $\varepsilon > 0$, there exists $\eta > 0$ such that if $|t - t'| < \eta$, then $d_\ell(f(t), f(t')) < \varepsilon$.

Let $a = t_0 \leq t_1 \leq \cdots \leq t_{n+1} = b$ be a partition of I having increments no larger than η. For each integer i between 0 and n, there exists a map g_i in $C([t_i, t_{i+1}])$ having the same values as f at t_i, t_{i+1} such that

$$\ell(g_i) \leq d_\ell(f(t_i), f(t_{i+1})) + \varepsilon/n.$$

By juxtaposing the g_i, we obtain a curve h_ε satisfying

$$\ell(h_\varepsilon) = \sum_{i=0}^{n} \ell(g_i) \leq \sum_{i=0}^{n} d_\ell(f(t_i), f(t_{i+1})) + \varepsilon \leq \tilde{\ell}(f) + \varepsilon$$

and such that for each $t \in I$, we have $d_\ell(h_\varepsilon(t), f(t)) \leq 3\varepsilon$.

From the hypothesis that ℓ is lower semicontinuous, it follows that

$$\ell(f) \leq \liminf_{\varepsilon \to 0} \ell(h_\varepsilon) \leq \tilde{\ell}(f),$$

whereas the opposite inequality is an immediate consequence of the definition of $\tilde{\ell}$.

Remark: If ℓ is the length structure associated with a metric d, then the same argument as above shows that $\ell = \tilde{\ell}$, using the semicontinuity of length with respect to d (cf. 1.2 and [Choq], p. 137). In other words, by following the sequence of constructions

(X, d), a metric space	$\longrightarrow$	metric length structure ℓ on X	$\longrightarrow$	(X, d_ℓ), a new metric associated with the length structure,

we obtain the same length structure. Nevertheless, we again emphasize that $\ell \neq \tilde{\ell}$ in general.

$1.6\frac{1}{2}_+$ Locality of the length structure. If two length structures agree on some open subsets covering X, then they are obviously equal. Conversely, if we are given a covering of X by open subsets X_i for $i \in I$, together with length structures ℓ_i on the X_i which are compatible on the intersections $X_i \cap X_j$ for all $i, j \in I$, then there obviously exists a (unique) length structure on X that restricts to ℓ_i on each X_i. (In other words, the length structures comprise a *sheaf* over X.) On the other hand, metrics on X are not local (they form only a presheaf over X), but they can be localized as follows: Given a metric d on X, we consider all metrics d' that are *locally majorized* by d. This means that for each $x \in X$, there exists a neighborhood $Y_x \subset X$ of x such that $d|_{Y_x} \geq d'|_{Y_x}$. Now take the supremum of all these d' and call it d_m. (Note that the supremum of a bounded family of metrics is again a metric. In general, this supremum may be infinite at some pairs of points in X, but otherwise it looks like a metric.) Clearly $d_m \leq d_\ell$ in any metric space (X, d); if (X, d) is *complete*, then $d_m = d_\ell$, as a trivial argument shows (see Section 1.8 below).

B. Path metric spaces

1.7. Definition: A metric space (X, d) is a *path metric space* if the distance between each pair of points equals the infimum of the lengths of curves joining the points (i.e., if $d = d_\ell$).

Examples: Note that, according to this definition the Euclidean plane is a path metric space, but the plane with a segment removed is *not*.

The n-sphere S^n is not a path metric space when equipped with the metric induced by that of $\mathbb{R}^{n+1}$, but it *is* a path metric space for the geodesic metric by Proposition 1.6.

Path metric spaces admit the following simple characterization.

1.8. Theorem: *The following properties of a metric space (X, d) are equiv-*

alent:

1. *For arbitrary points $x, y \in X$ and $\varepsilon > 0$, there is a z such that*

$$\sup(d(x,z), d(z,y)) \leq \frac{1}{2}\, d(x,y) + \varepsilon.$$

2. *For arbitrary $x, y \in X$ and $r_1, r_2 > 0$ with $r_1 + r_2 \leq d(x,y)$, we have*

$$d(B(x,r_1), B(y,r_2)) \leq d(x,y) - r_1 - r_2,$$

for

$$d(B_1, B_2) = \inf_{\substack{x' \in B_1 \\ y' \in B_2}} d(x', y').$$

Every path metric space has these properties, and conversely, if (X,d) is complete and satisfies (1) or (2), then it is a path metric space.

Proof. Let (X,d) be a complete metric space satisfying condition (1), and set $\delta = d(x,y)$. Given a sequence (ε_k) of positive numbers, there is a point $z_{1/2}$ such that $\max(d(x,z_{1/2}), d(z_{1/2},y)) \leq \delta/2 + \varepsilon_1\delta/2$, and points $z_{1/4}, z_{3/4}$ for which each of the distances $d(x,z_{1/4}), d(z_{1/4},z_{1/2}), d(z_{1/2},z_{3/4}), d(z_{3/4},y)$ are less than

$$1/2(\delta/2 + \varepsilon_1\delta/2) + \varepsilon_2(\delta/2 + \varepsilon_1\delta/2), \text{ etc.}$$

By choosing the sequence (ε_k) so that $\sum_k \varepsilon_k < \infty$, we can define a map f from the dyadic rationals in $[0,1]$ into X satisfying

$$d\left(f\left(\frac{p}{2^n}\right), f\left(\frac{p+1}{2^n}\right)\right) \leq \frac{\delta}{2^n} \prod_{k=1}^{\infty} (1 + \varepsilon_k).$$

If (X,d) is complete, then this map extends to the entire interval $[0,1]$. Since we can choose the ε_k so that the product $\prod(1+\varepsilon_k)$ is arbitrarily close to 1, we obtain curves whose lengths tend to $\delta = d(x,y)$, which proves the last assertion.

The implication (1) $\Rightarrow$ (2) is proven in the same way, whereas (2) $\Rightarrow$ (1) and the assertion that a path metric space satisfies (1),(2) are trivial.

Path metric spaces enjoy some of the same geometric properties as Riemannian manifolds.

1.8bis. Property: If (X,d) is a path metric space, and if f is a map of X into a metric space Y, then the dilatation of f obviously equals the supremum of its local dilatation, i.e., $\mathrm{dil}(f) = \sup_{x \in X} \mathrm{dil}_x(f)$. Note that if X and Y are Riemannian manifolds, and if f is differentiable, then the differential $Df_x: T_xX \to T_{f(x)}Y$ satisfies $\mathrm{dil}_x(f) = \|Df_x\|$.

1.8bis$_+$ Kobayashi metrics. Let $\triangle$ be a path metric space and let X be an arbitrary (say, topological) space with a distinguished set of maps $f: \triangle \to X$. Consider all metrics d' on X for which these f have $\mathrm{dil}(f) \leq 1$, i.e., for which the mappings are (nonstrictly) distance decreasing, and define d_K as the supremum of the metrics d' on X. Here, it is convenient to admit *degenerate* metrics d' (in the sense that $d'(x, y) = 0$ for perhaps some $x \neq y$), so that d_K may itself be degenerate. In fact, this d_K is a (possibly degenerate) path metric by the property above.

In the classical example, due to Kobayashi, $\triangle$ is the unit open disk equipped with the Poincaré metric (i.e., the hyperbolic plane), X is a complex analytic space, and the collection of distinguished maps consists of all holomorphic mappings $\triangle \to X$. The usefulness of this metric is based on the *Schwarz lemma* (and its various generalizations), which implies that d_K is *nondegenerate* for many X. Such X are said to be (Kobayashi) *hyperbolic*. For example, the disk $\triangle$ is itself hyperbolic since d_K in this case equals the Poincaré metric (following from the fact that every holomorphic map $\triangle \to \triangle$ is distance decreasing with respect to the Poincaré metric, a consequence of the classical Schwarz-Ahlfors lemma). The basic features of the Kobayashi metric and hyperbolicity do not depend on the integrability of the implied (almost) complex structure of X and therefore extend to all *almost complex* manifolds X (via the theory of pseudo-holomorphic curves in X, cf. [McD–Sal]). For example, hyperbolicity is stable under small (possibly singular) perturbations of almost complex structures on compact manifolds and (suitably defined) singular almost complex spaces (compare [Kobay], [Brody], [Krug–Over]).

There is also a real analog of Kobayashi hyperbolicity, in which X is a Riemannian manifold, $\triangle$ is as above, and the set of distinguished maps consists of all conformal, globally area minimizing mappings $\triangle \to X$. In this case, hyperbolicity of X is equivalent to δ-hyperbolicity (see (e) in 1.19_+ below) under mild restrictions on X, which are satisfied, for example, if X is the universal cover of a compact manifold. (In fact, X does not have to be a manifold here — it can be a rather singular space, e.g., a simplicial polyhedron as in 1.15_+, see $[\mathrm{Gro}]_{\mathrm{HG}}$, $[\mathrm{Gro}]_{\mathrm{HMGA}}$.)

1.9. Definition: A *minimizing geodesic* in a path metric space (X, d) is any curve $f: I \to X$ such that $d(f(t), f(t')) = |t - t'|$ for each $t, t' \in I$. A *geodesic* in X is any curve $f: I \to X$ whose restriction to any sufficiently small subinterval in I is a minimizing geodesic.

In this connection, we have the following :

Hopf–Rinow theorem. *If (X, d) is a complete, locally compact path metric space, then*

1. *Closed balls are compact, or, equivalently, each bounded, closed domain is compact.*

2. *Each pair of points can be joined by a minimizing geodesic.*

Before turning to the proof of the theorem, we observe that if (X, d) is a complete, locally compact metric space, then there are many noncompact balls for the metric $d' = \inf(1, d)$.

1.11 Compactness of closed balls. Note that if a is a point in X, then the ball $B(a, r)$ is by hypothesis compact if r is sufficiently small. We will first show that if $B(a, r)$ is compact for all r in an interval $[0, \rho)$, then $B(a, \rho)$ is compact as well.

Let (x_n) be a sequence of points in $B(a, \rho)$. We may suppose that the distances $d(a, x_n)$ tend to ρ; otherwise, there is a ball $B(a, r)$ with $r < \rho$ containing infinitely many of the x_n and thus a limit point of the sequence. Let (ε_p) be a sequence of positive real numbers tending to zero. By applying property (2) of Theorem 1.8, we find that for each p, there exists an integer $n(p)$ such that for each $n > n(p)$, there is a point y_n^p satisfying

$$y_n^p \in B(a, \rho - 2\varepsilon_p) \qquad \text{and} \qquad d(x_n, y_n^p) \leq \varepsilon_p.$$

For each p, the sequence (y_n^p) lies within a compact set; by a diagonal argument (or since the product of compact sets is compact), it follows that there is a sequence of integers (n_k) such that the subsequence $(y_{n_k}^p)$ converges for all p. The sequence (x_{n_k}), which is the uniform limit of the $(y_{n_k}^p)$, is a Cauchy sequence and therefore converges by completeness of X.

By the preceding remarks, the supremum of the r for which $B(a, r)$ is compact is infinite: if instead it equalled $\rho < \infty$, then we could find $\rho' > \rho$ such that $B(a, \rho)$ would be compact, by using a finite covering of the sphere $S(a, \rho)$ by compact balls.

1.12. Existence of a minimizing geodesic joining two arbitrary points

We first consider the case when X is compact.

Lemma: *If (X, d) is a compact path metric space and $a, b \in X$, then there exists a curve of length $d(a, b)$ joining a and b.*

Proof. It suffices to consider curves $f\colon [0,1] \to X$ which are parametrized by arc length. From the definition of path metric spaces, if follows that for each positive integer n, there exists such a curve f_n joining a to b and having length less than $d(a,b) + 1/n$. The set of f_n is therefore equicontinuous, and by Ascoli's theorem, there exists a subsequence f_{n_k} that converges uniformly to a curve $f : [0,1] \to X$. Since the length function ℓ is lower semicontinuous, we have

$$\ell(f) \leq \liminf_{k\to\infty} \ell(f_{n_k}) = d(a,b).$$

In the case of a complete, locally compact, but noncompact path metric space, it suffices to note that the images of the curves f_n chosen in the preceding paragraph lie within the compact ball $B(a, 2d(a,b))$.

1.13. Remarks: (a) In the case of Riemannian manifolds, this proof fulfills the promise made in the introduction that use would be made only of the associated length structure.

(b) The equicontinuity argument of Lemma 1.12 also shows that *in a compact path metric space, every free homotopy class is represented by a length-minimizing curve*, and that the minimizing curves are geodesics. Moreover, if X is a manifold, then for each real r, *there is only a finite number of homotopy classes represented by curves of length less than* r (again, it suffices to use Ascoli's theorem and the fact that the homotopy classes are open subsets of $C^0(S^1, X)$; cf. [Dieu], p. 188). These results also hold for homotopy classes of curves based at a point $x \in X$ and geodesics based at x (but not necessarily smooth at x) and will play a key role, particularly in Chapter 5.

C. Examples of path metric spaces

1.14. Riemannian manifolds with boundary and subsets of $\mathbb{R}^n$ with smooth boundary. Let X be a domain in $\mathbb{R}^n$ with smooth boundary, equipped with the metric and length structure induced by that of $\mathbb{R}^n$, and let f be the identity map

$$(X, \text{induced metric}) \to (X, \text{induced length metric}).$$

It is easy to see that if the boundary of X is smooth, then $\mathrm{dil}_x(f) = 1$ for each $x \in X$, and that $\mathrm{dil}(f) = 1$ if and only if X is convex.

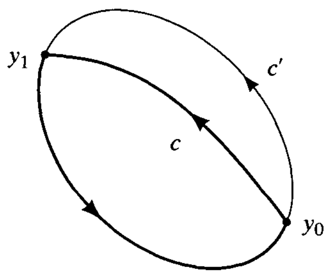
Distortion$_+$: More generally, let X be a subset of a path metric space A and let distort(X) denote the dilatation of the identity map $f\colon X \to X$ with respect to the two induced metrics, i.e.,

$$\operatorname{distort}(X) = \sup \frac{(\text{length dist})|_X}{\text{dist}|_X}.$$

Our first observation is the following:

(a) Let X be a compact subset of $\mathbb{R}^n$. If distort$(X) < \frac{\pi}{2}$ *(which means that every two points in X that lie within a Euclidean distance of d from one another can be joined by a curve in X of length $< d\pi/2$), then X is simply connected.*

Proof. To prove this assertion, we argue by contradiction. Suppose $\pi_1(X) \neq 0$ and let α be a nontrivial homotopy class in which there exists a curve of minimal length among all homotopically nontrivial loops (the existence of such α is guaranteed by Remark 1.13(b)). Let Y be the image of c and $g : Y \to Y$ the identity map of the space Y with its induced length structure. We claim that $\operatorname{dil}(g) = \operatorname{dil}(f|_Y)$.



To prove the claim, let y_1, y_0 be two points of Y and fix a parametrization of Y by arc length, i.e., a map $c\colon [0,\ell] \to Y$ such that $c(0) = y_0 = c(\ell)$ and $y_1 = c(d)$ for some $d \in [0,\ell]$ such that $d \leq \ell - d$. Then $c|_{[0,d]}$ is the shortest path joining y_0 to y_1 in X. Indeed, if there were a strictly shorter path c' from y_0 to y_1, then the two loops obtained by adjoining c' and the two parts of c defined by the parameters 0 and d would be strictly shorter than c. Since their product is homotopic to c, however, we could conclude that one of the two is not homotopic to 0 in X, which contradicts the minimality of c. Since the path $c|_{[0,d]}$ lies within Y, it follows that d is the distance from y_0 to y_1 for the length metrics of X and Y, and that

$$\operatorname{dil}_{(y_0,y_1)}(g) = \operatorname{dil}_{(y_0,y_1)}(f).$$

Thus, we have $\mathrm{dil}(g) < \pi/2$. Extend c to a periodic function on $\mathbb{R}$ and set $r(s) = d(c(s), c(s+\ell/2))$, so that the inequality $r(s) \geq \ell/2\,\mathrm{dil}(g)$ holds. Set $u(s) = (c(s+\ell/2) - c(s))/r(s)$. The curve u is differentiable almost everywhere, and its image lies within the unit sphere of $\mathbb{R}^n$. Moreover, $u(s+\ell/2) = -u(s)$, so that the length of u is at least 2π. Thus,

$$\left\|\frac{du}{ds}\right\|^2 \leq \frac{1}{r(s)^2}\left(4 - \left(\frac{dr}{ds}\right)^2\right) \leq \frac{4}{r(s)^2} \leq \left(\frac{4\,\mathrm{dil}(g)}{\ell}\right)^2,$$

and so $\ell(u) \leq 4\,\mathrm{dil}(g) < 2\pi$, which is the desired contradiction.

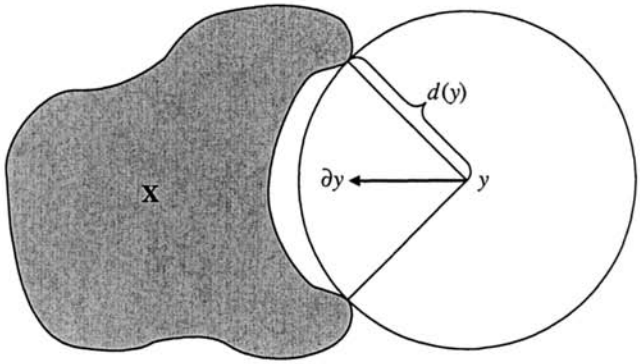
Remark: If $\mathrm{dil}(d) = \pi/2$, and if X is not simply connected, then X contains a round circle.

(*b*) *If* $\mathrm{distort}(X) < \pi/2\sqrt{2}$, *then X is contractible.*

Proof$_+$. The idea is to homotopy-retract $\mathbb{R}^n$ to X by following the flow of a suitable vector field ∂ which plays the role of $-\operatorname{grad} d(y)$ for the distance function $d\colon y \mapsto \mathrm{dist}(y, X) = \inf_{x\in X} \|y - x\|$. In general, the function d is nonsmooth, even on the complement $\mathbb{R}^n \setminus X$. Nonsmoothness at a point $y \in \mathbb{R}^n \setminus X$ is due to the fact that the sphere S_y^{n-1} at y of radius $d(y)$ can meet X at several points. These points essentially realize the above infimum, since the *open* ball bounded by S_y^{n-1} does not intersect X, while the set $X \cap S_y^{n-1}$ is nonempty and $d(y) = \|x - y\|$ for all $x \in X \cap S_y^{n-1}$. Now we observe that the normal projection $X \to S_y^{n-1}$ is distance-decreasing and thus $(\text{length dist})_X(x_1, x_2) \geq (\text{length dist})_{S_y^{n-1}}(x_1, x_2)$ for all pairs of points in the intersection $X \cap S_y^{n-1}$. It follows that the latter distance is bounded by $\pi d(y)/2$, and since we assume the *strict* inequality $\mathrm{distort}(X) < \pi/2\sqrt{2}$, the distance above is bounded by $\delta\pi d(y)/2$ for some $\delta < 1$ independent of y. Consequently, the intersection $X \cap S_y^{n-1}$ is *strictly* contained in a hemisphere, or, in other words, *there exists a unit vector ∂_y at every $y \in \mathbb{R}^n \setminus X$ such that*

$$\partial_y \|x - y\| \leq -\varepsilon < 0 \qquad \textit{for all} \qquad x \in X \cap S_y^{n-1}, \tag{$*$}$$

where $\partial_y\|x - y\|$ denotes the ∂_y-derivative of the (distance) function $y \mapsto \|x - y\|$. In fact, one can take ∂_y to be the vector which points towards center of the minimal spherical cap in S_y^{n-1} containing $X \cap S_y^{n-1}$, thus obtaining a (Borel) measurable vector field $y \mapsto \partial_y$ satisfying $(*)$.
Finally, we can easily smooth this vector field, so that the resulting (now smooth!) unit vector field, say $y \mapsto \overline{\partial}_y$ on $\mathbb{R}^n \setminus X$, satisfies $(*)$ (possibly with a smaller $\varepsilon > 0$) as well. Clearly, every forward orbit of such a field



converges to a point in X, and so the flow generated by $\overline{\partial}$ eventually retracts $\mathbb{R}^n$ to X.

Remark$_+$: A sharper result is proved in Appendix A, where one allows $\text{distort}(X) < 1+\alpha_n$ for some specific $\alpha_n > 1-\pi/2\sqrt{2}$. Furthermore, there are many examples of (necessarily contractible) subsets X with arbitrarily large (even infinite) distortion for which $X \cap S_y^{n-1}$ is still strictly contained in an open hemisphere for each $y \in \mathbb{R}^n \setminus X$ and to which our argument applies. On the other hand, we do not know the precise value of α_n for which $\text{distort}(X) < 1+\alpha_n$ necessarily implies that X is contractible.

Exercise$_+$: Construct a closed, convex surface X in $\mathbb{R}^3$ with $\text{distort}(X) < \pi/2$. (Compare Appendix A.)

Problem$_+$: Given a topological space X, evaluate the infimum of all distortions induced by embeddings $X \to \mathbb{R}^n$ or of those distortions induced by embeddings which lie in a fixed isotopy class. The first interesting case arises when X is the circle and we minimize the distortion for X knotted in $\mathbb{R}^3$ in a prescribed way (compare [Gro]$_{\text{HED}}$, [O'Hara]$_{\text{EK}}$).

Remark$_+$: The geometry of subsets $X \subset \mathbb{R}^n$ satisfying $\text{distort}(X) \le 1+\alpha_n$ can be rather complicated, even for small $\alpha_n > 0$. For example, there are simple smooth arcs in $\mathbb{R}^2$ with arbitrarily small (i.e. close to 1) distortion which have an arbitrarily large turn of the tangent direction. To see this, consider diffeomorphisms $T_i\colon \mathbb{R}^2 \to \mathbb{R}^2$ with the following properties: Each T_i fixes the complement of the disk of radius 2^{-i} around the origin and isometrically maps the disk of radius 2^{-i-1} into itself by rotating it by a small angle $\alpha > 0$.

Clearly, one can choose the T_i so that they and their inverses are $(1+\varepsilon)$-